

## ❖ 45 Days Summer Training Program

### ➤ Assignment: 1

1. Explain the difference between the following data types with examples:

- Integer (int) :-

- An integer is a whole number that can be positive, negative, or zero.

- Example :-

```
x = 25
print(x)
print(type(x))
```

- Output :-

```
25
<class 'int'>
```

- Float (float) :-

- float data type use to represent real numbers that contain a decimal point or fractional component.

- Example :-

```
y = 3.14
print(y)
print(type(y))
```

- Output :-

```
3.14
<class 'float'>
```

- String (str) :-

- a string is an immutable sequence of characters used to represent text data.

- Example :-

```
name = "ABC"
print(name)
print(type(name))
```

➤ Output :-

```
ABC
<class 'str'>
```

• Boolean (bool) :-

➤ Boolean is a built-in data type that represents one of two possible values. True or False.

➤ Example:-

```
is_raining = False
print(is_raining)
print(type(is_raining))
```

➤ Output:

```
False
<class 'bool'>
```

2. Write a Python program to create three variables:

- name
- age
- city

Store your details in them and print the values.

➤ Python program :-

```
name = "Pankaj"
age = 18
city = "JNN"
print("Name:", name)

print("Age:", age)

print("City:", city)
```

➤ Output :-

```
Name: Pankaj
Age: 18
City: JNN
```

3. Write a Python program that:  
Takes a user's name as input.  
Prints the name in uppercase.  
Prints the total number of characters in the name.

➤ Python program :-

```
name = input("Enter your name: ")  
print("Uppercase Name:", name.upper())  
print("Total Characters:", len(name))
```

➤ Output :-

```
Enter your name: Pankaj  
Uppercase Name: PANKAJ  
Total Characters: 6
```

4. Explain any five commonly used string methods in Python with examples.

I. upper() :-

Converts all characters in a string to uppercase.

a) Example :-

```
name = "pankaj"  
print(name.upper())
```

b) Output :-

```
PANKAJ
```

II. lower() :-

Converts all characters in a string to lowercase.

c) Example :-

```
text = "HELLO"  
print(text.lower())
```

d) Output :-

```
hello
```

III. strip() :-

Removes leading and trailing spaces from a string.

e) Example :-

```
text = " Python "  
print(text.strip())
```

f) Output :-

```
Python
```

IV. replace() :-

Replaces a specified word or character with another.

g) Example :-

```
text = "I like Java"
print(text.replace("Java", "Python"))
```

h) Output:-

I like Python

V. split() :-

Splits a string into a list based on a separator (default is space).

i) Example :-

```
text = "Apple Banana Mango"
print(text.split())
```

j) Output :-

['Apple', 'Banana', 'Mango']

5. Create a list containing the names of five fruits.

- Print the complete list.
- Print the first and last element.
- Print the total number of items in the list.

➤ Code:-

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]
print("Complete List:", fruits)
print("First Element:", fruits[0])
print("Last Element:", fruits[-1])
print("Total Number of Items:", len(fruits))
```

➤ Output:-

```
Complete List: ['Apple', 'Banana', 'Mango', 'Orange', 'Grapes']
First Element: Apple
Last Element: Grapes
Total Number of Items: 5
```

6. Write a Python program to:

- Create a list of numbers [10, 20, 30, 40, 50]
- Add 60 to the list.

- Remove 20 from the list.
- Print the updated list.

➤ Code :-

```
numbers = [10, 20, 30, 40, 50]
numbers.append(60)
numbers.remove(20)
print("Updated List:", numbers)
```

➤ Output :-

Updated List: [10, 30, 40, 50, 60]

7. What is Artificial Intelligence (AI)? Explain its importance and mention any four real-life applications of AI.

- Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines and software that can perform tasks that normally require human intelligence.

➤ Importance of AI :-

AI is important because it:

- Automates repetitive tasks and reduces human effort.
- **Increases** efficiency and productivity by performing tasks quickly and accurately.
- Helps in decision-making by analyzing large amounts of data.
- Reduces human errors in many processes.
- Provides 24/7 service through systems such as chatbots and virtual assistants.
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➤ Real-Life Applications of AI

1. Virtual Assistants :-

- AI powers virtual assistants such as Siri and Google Assistant, which can answer questions, set reminders, and perform tasks using voice commands.

## 2. Healthcare

- AI is used to help doctors diagnose diseases, analyze medical reports, and recommend treatments, leading to faster and more accurate healthcare services.

## 3. Online Shopping and Recommendations

- E-commerce websites use AI to recommend products based on a user's interests and previous purchases, making shopping more personalized.

## 4. Self-Driving Vehicles

- AI helps autonomous vehicles recognize roads, traffic signs, pedestrians, and other vehicles so they can navigate safely.

## 8. Identify whether the following are examples of AI and explain why:

- ChatGPT
- Google Maps route prediction
- Calculator
- Netflix recommendations
- Voice assistants (Alexa/Siri)

### ➤ ChatGPT :-

ChatGPT is an AI because it can understand what people type and respond in a natural way.

### ➤ Google Maps Route Prediction :-

Google Maps uses AI to analyze traffic, road conditions, and travel history. It can predict the fastest route and estimate how long a trip will take.

### ➤ Calculator :-

A calculator is not AI because it only follows fixed mathematical rules. It cannot learn, think, or make decisions on its own. It simply performs calculations when instructed.

### ➤ Netflix Recommendations :-

Netflix uses AI to suggest movies and TV shows based on what you have watched before. It learns your preferences and recommends content you are likely to enjoy.

➤ Voice Assistants (Alexa/Siri) :-

Alexa and Siri are AI-powered assistants. They can understand spoken commands, answer questions, play music, set alarms, and perform other tasks by processing human language.

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